

Database for LLM

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Sep. 15th, 2026

Research Topics change

VLDB 2022

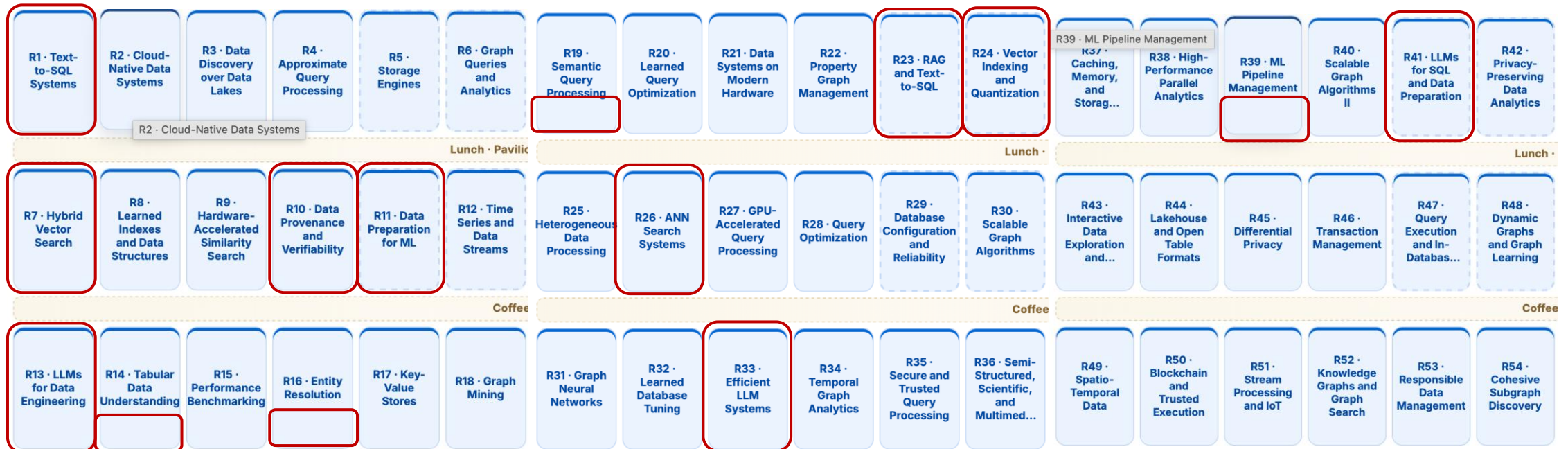
Database Engines 1-10 // Specialized and Domain-Specific Data Mgmt 1-3

ML, AI and DB 1-8 // **Graphs 1-9** // Novel DB Architectures 1 // Data Mining 1-2 // Distributed DB systems 1 //

Data Mining and Analytics 1-3 // Information Integration // Information Integration & Data Quality 1-2 //

Data Privacy and Security 1-2 // Database Performance 1-2

VLDB 2026 Topics and Sessions



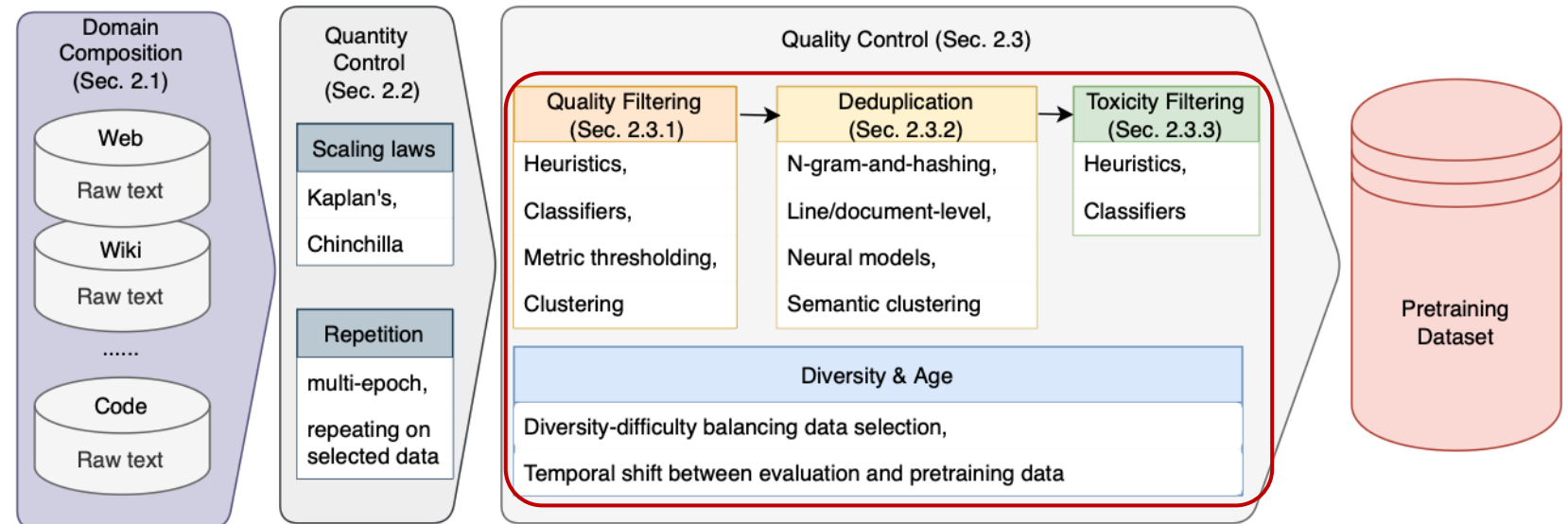
Recent research topics: LLM and DB

- Pre-training

- Open models (**Base** – *Instruct*)

- Qwen2.5 (Alibaba) – 0.5B, 1.5B, 7B, 14B, 32B, 72B
 - Llama 3.1 (Meta) – 8B, 70B, 405B
 - Gemma 3 (Google) – 1B, 4B, 12B, 27B
 - - 4B: 16GB // 7-8B: 18-24GB // 12-14B: 36GB (RTX 4090 24GB)

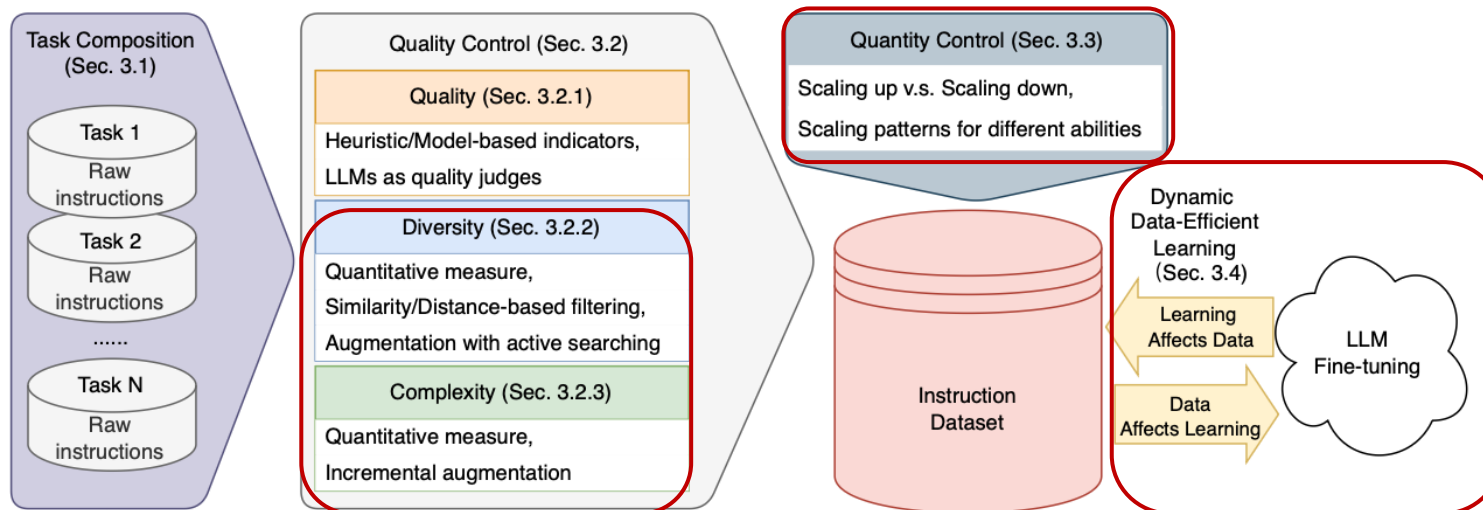
- **Data Management** in the pretraining
 - Quality control
 - Data curation



(a) Data management pipeline in the pretraining stage of LLMs

Recent research topics: LLM and DB

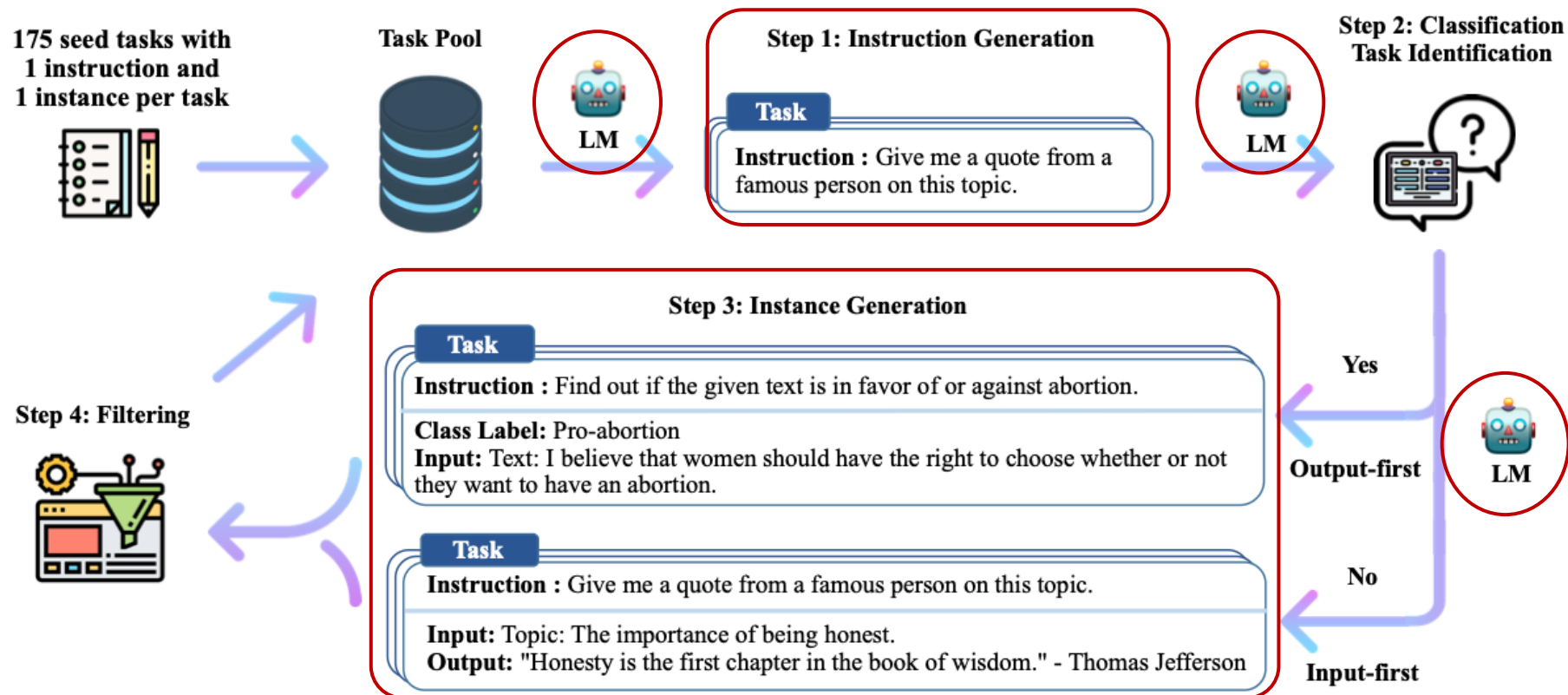
- Fine Tuning
 - Supervised Fine Tuning (SFT)
 - Labeled data (Instructions paired with desired responses)
 - Task adaptation, Human-in-the-loop learning
 - LoRA (Low-Rank Adaptation) / PEFT (Parameter Efficient Fine Tuning)
 - Update a small set of parameters to reduce training cost
 - Parameter efficient and keeping its original weights ($W' = W + BA$)



(b) Data management pipeline in the supervised fine-tuning stage of LLMs

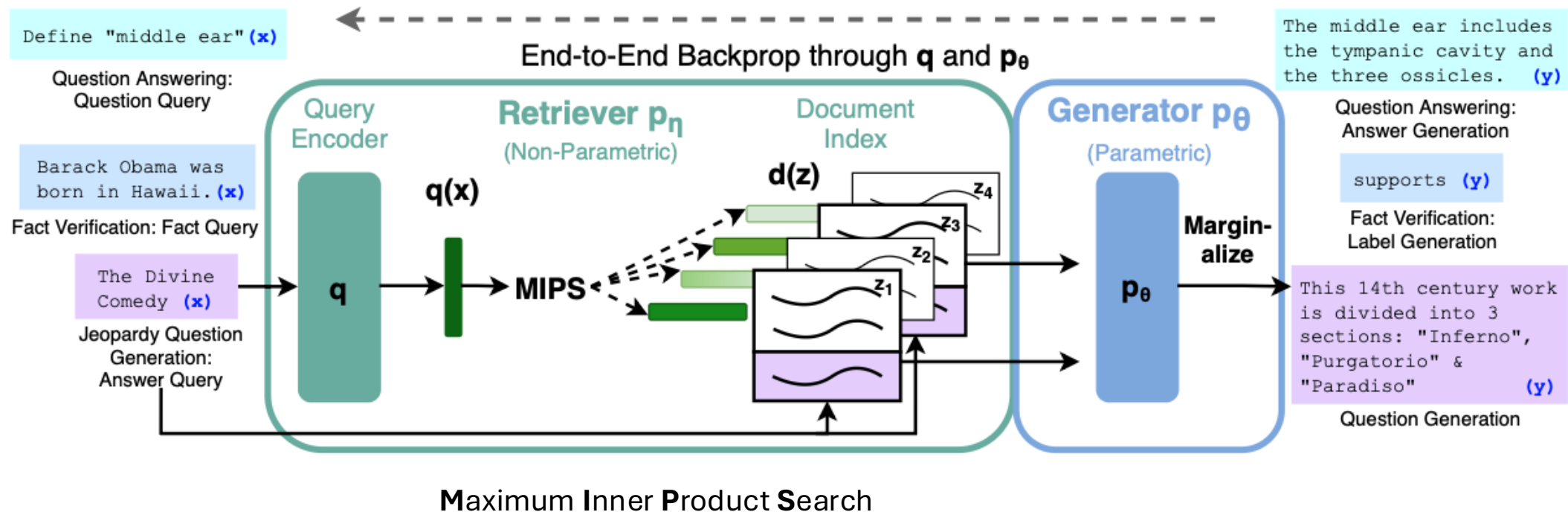
Self-Instruct (ACL 2023)

- Finetuned to respond to instruction
 - From human-written → LLM-written (**self-instruction**)



Recent research topics: LLM and DB

- RAG (Retrieval Augmented Generation)
 - Augments the input context during inference
 - Retrieval (ex. Dense Passage Retrieval) – Generation (seq-to-seq)



Research keywords in RAG

keyword	Meaning
Retrieval (general)	Finding relevant documents , passages, or table data
Embeddings	Vector representations used for semantic search (ex. Documents)
Chunking	Splitting documents into units for retrieval
Hybrid search	Combining keyword-based and semantic retrieval
Reranking	Reordering retrieved results by relevance
Retrieval evaluation	Measuring retrieval quality , e.g., Recall@k
Noise robustness	Handling irrelevant, incorrect, or conflicting evidence (ex. avoid hallucination, answering unknown, ...)
Tabular RAG	Retrieving and reasoning over structured tables

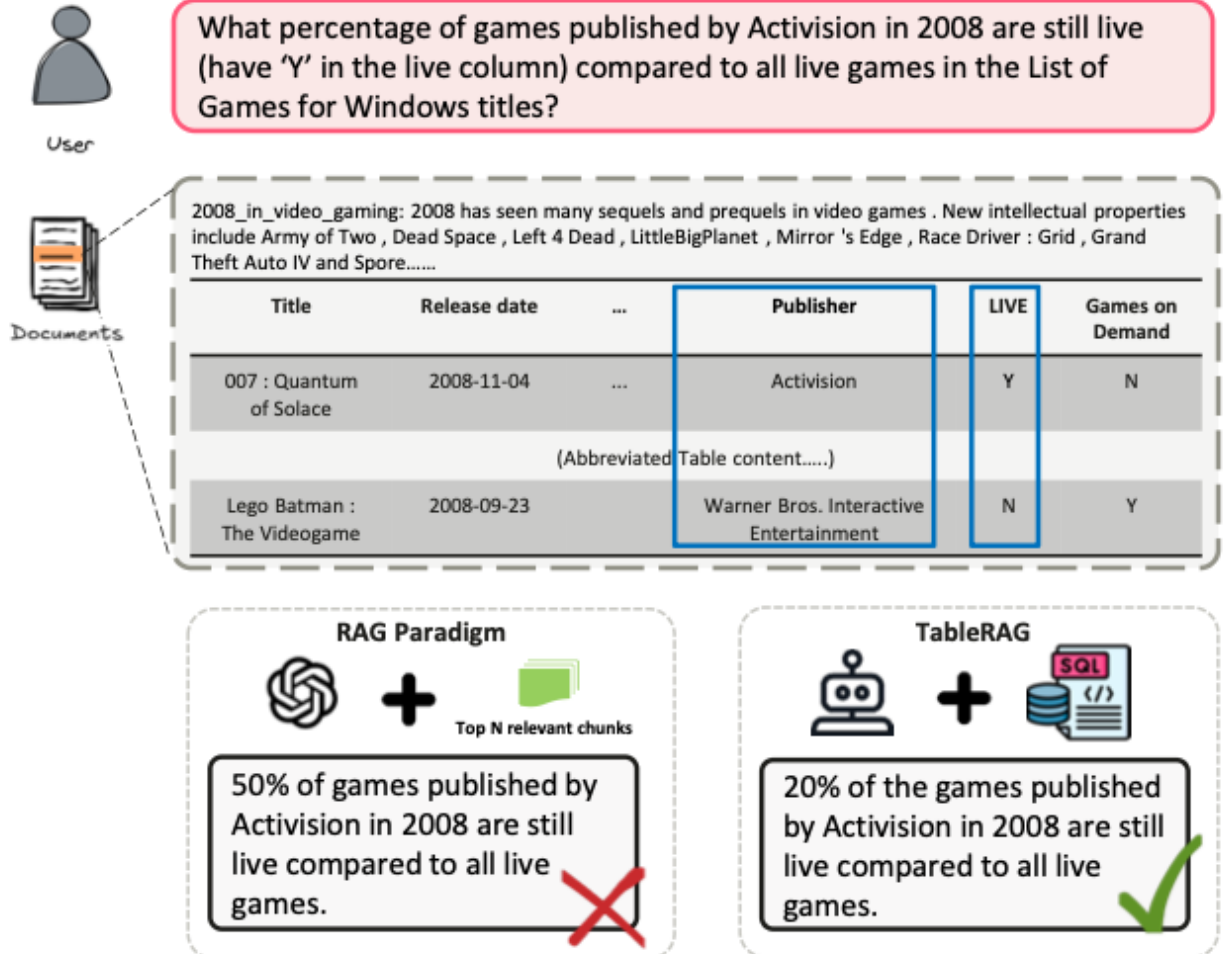
Vector DB and data quality

- Vector database
 - Embedding documents and use it for retrieval
- Approximate Nearest Neighbor (ANN)
 - Accelerate searching

Research Area / Keyword	Key Research Question
Index Design	Graph - or cluster-based indexes, efficiency / nearest
Recall–Latency Trade-off	Retrieval latency vs. high recall
Compression / Quantization	Vector compression, reduce memory & retrieval quality
Filtered ANN	Efficient similarity search w/ dates, categories, and access permissions
Dynamic Indexing	Searching quality and speed for insertion, updating, or deletion
Scalable Search	Managements of SSDs, GPUs, and distributed systems

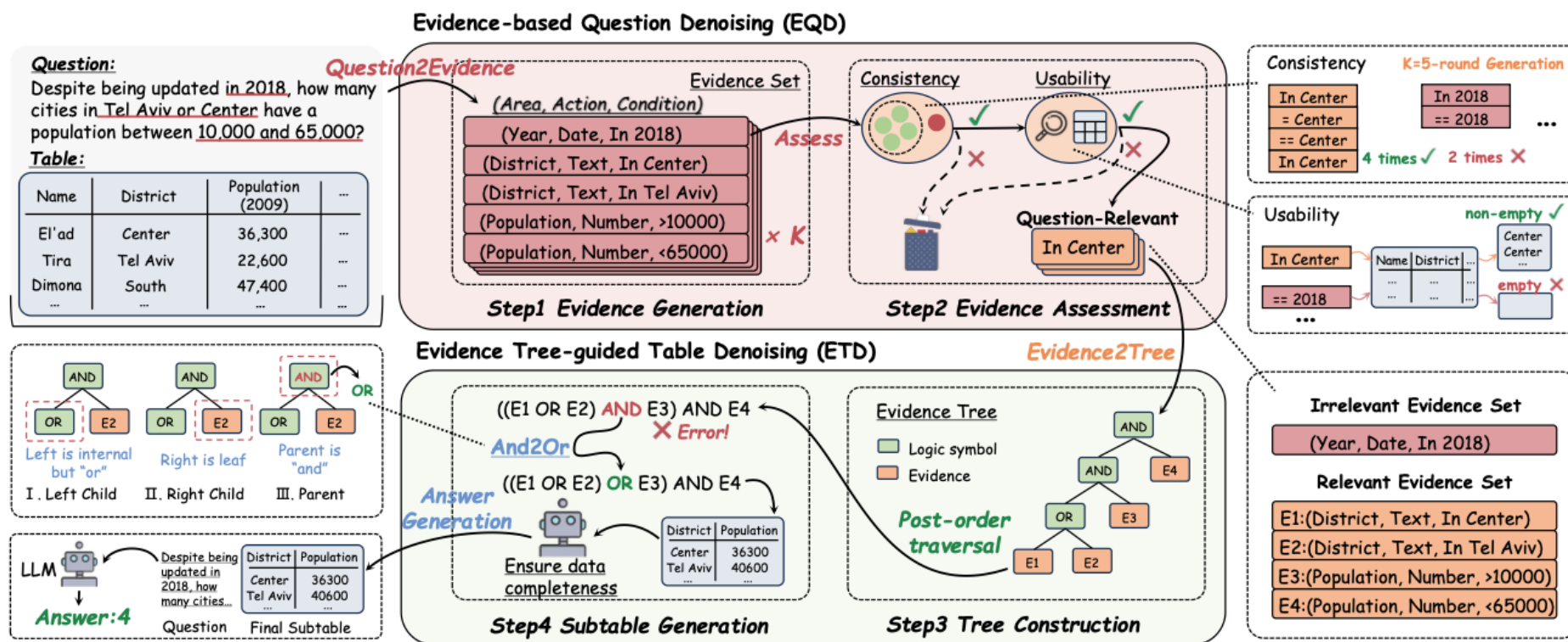
TableRAG (EMNLP 2025)

- Heterogeneous documents (text and tabular)
 - **SQL programming** and execution for tabular data



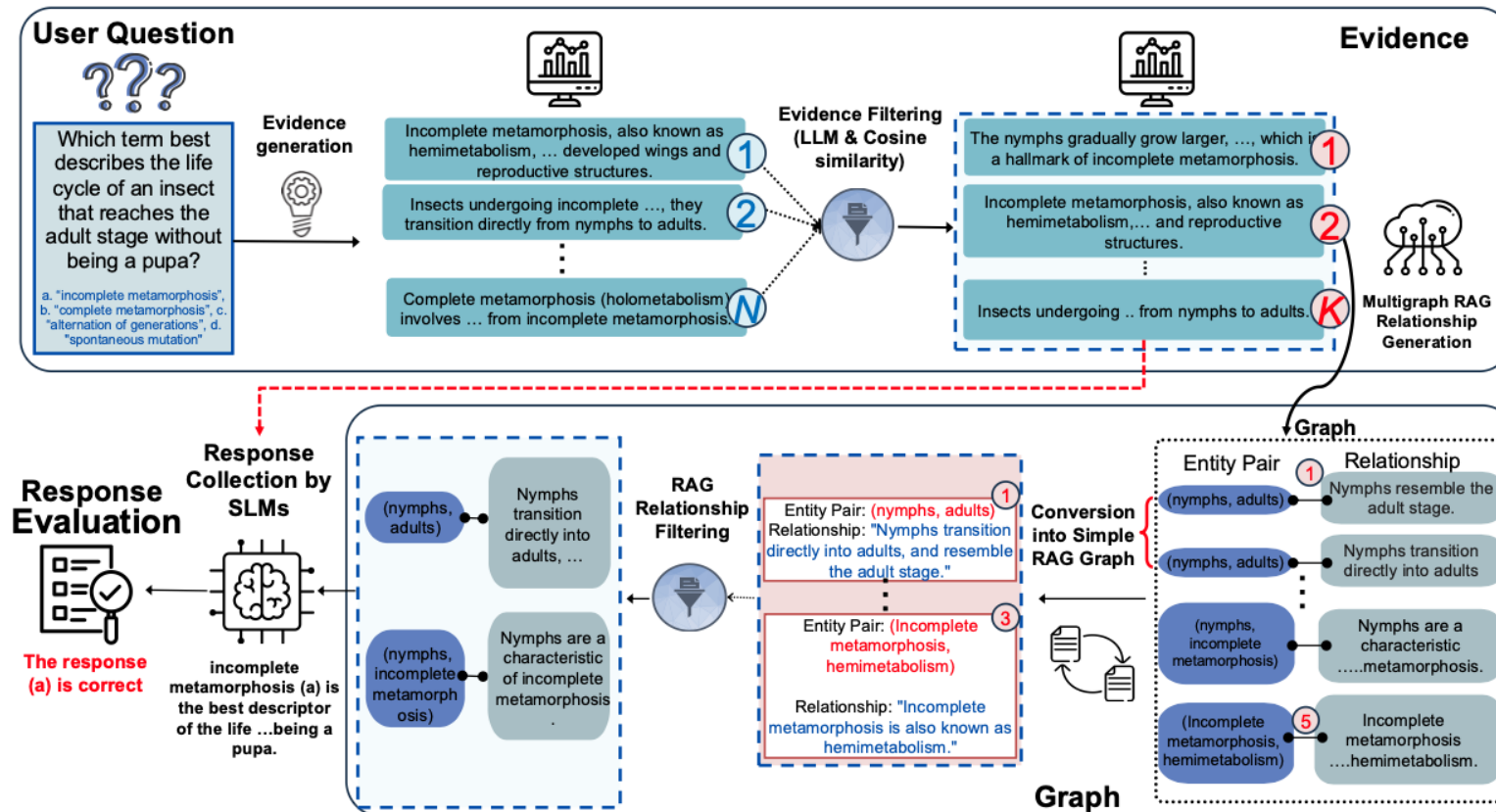
EnoTab (ACL 2026)

- Complex questions and larger tables → noisy data
 - Relevance Filtering: retain information
 - Table Pruning: reduce table size



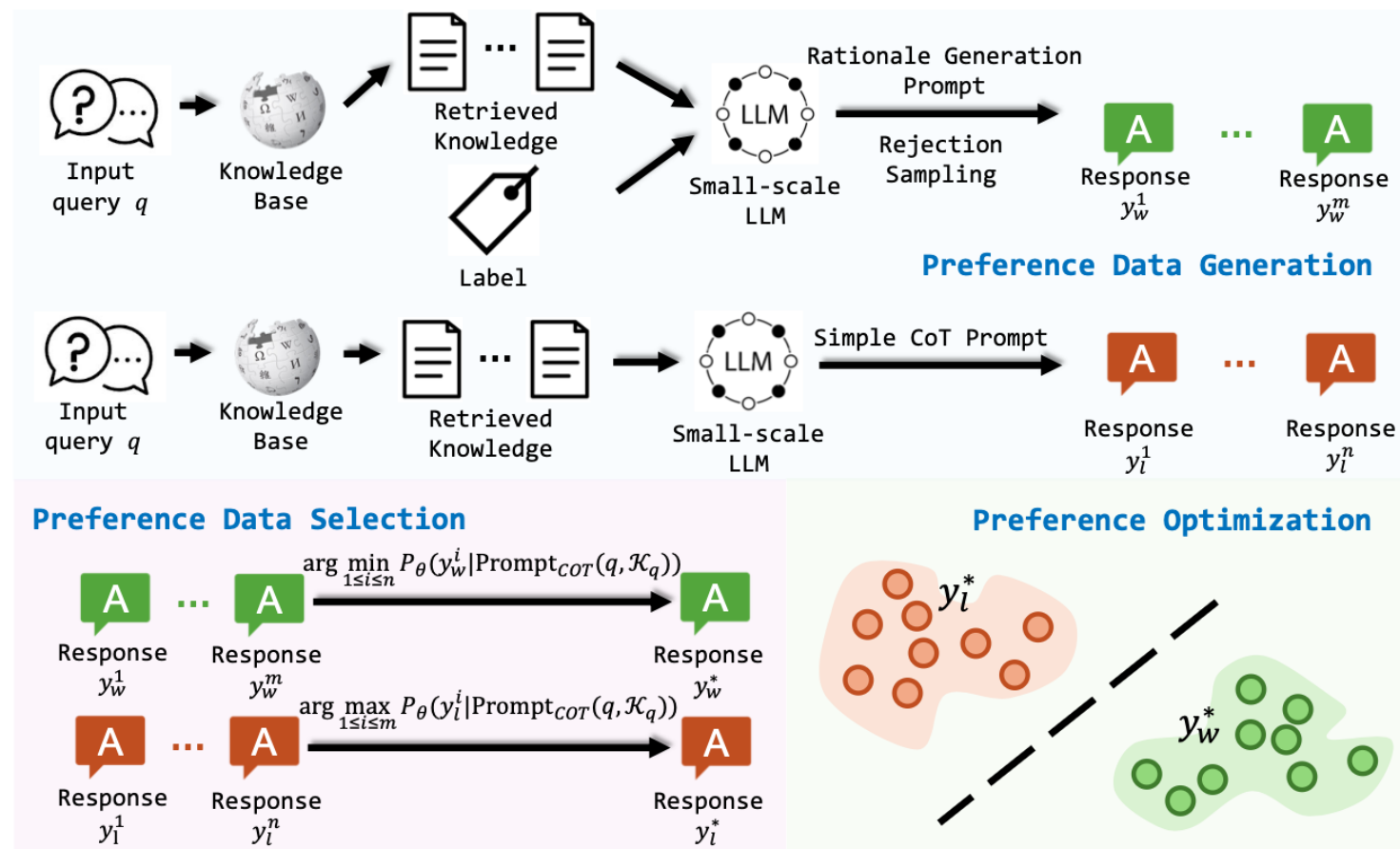
DRAG (ACL 2025)

- LLM to SLM: Transfer knowledge and mitigate hallucination
- Graph-based distillation



ROSERAG (ACL 2025)

- For SLM
 - Preferred data selection for training
 - Multi-turn prompting
 - Rejection sampling
 - Contrastive preference selection



Database and LLM: Research

Area	Role	Topics
RAG (Retrieval-Augmented Generation)	Retrieve data to support evidence-grounded answers	• Data discovery and integration • Evidence relevance and sufficiency • Data quality, provenance
ANN (Approximate Nearest Neighbor)	Efficiently find similar vectors in large datasets	• Index design • Recall–latency trade-offs • Filtered search, compression • Dynamic updates
Text-to-SQL	Translate natural-language questions into executable SQL	• Schema linking • Join-path selection • Ambiguity resolution • SQL correctness verification

Plan

- Not limited to these topics
- Will open github (lab)
 - Research topics / paper lists ...
 - Choose some of them or add topics / paper lists ...
 - No pressure, but prepare short introduction / presentation
 - Ask to read papers together and discuss
 - Of course, we can share ideas and prepare paper works